

Product Management for Physical + Digital Systems · Session Reading List

Core readings: 1–3 per session, selected for practical skills and case grounding · Graduate / Engineering: 6–8 readings, full technical and strategic apparatus

WK	TOPIC & SESSION FOCUS	UNDERGRADUATE / MBA (1–3 readings) <i>Core skills focus</i>	GRADUATE / ENGINEERING (6–8 readings) <i>Full technical and strategic apparatus</i>
1	Why Hardware Is Different <i>The atom-bit divide: physical constraints, irreversibility, and why the standard software PM playbook breaks on contact with the real world.</i>	- Cagan, Marty, Inspired, Ch. 1–3 & 7 (Key Concepts) (2018) — baseline PM vocabulary — read for what hardware violates - Mocker, Monica & Arnold, David, When Your Business Model Is Your Moat: The Case of Nest (2017) — HBS Case 9-617-051 — the canonical physical-digital product - Hoffman, Patrick T., @arcktip Practitioner Framework: Systems Thinking for Hardware PMs (2026) — @arcktip — the opening framing essay for the course	- Cagan, Marty, Inspired, Ch. 1–7 (2018) — Wiley — read fully; annotate where hardware violates the software assumptions - Mocker, Monica & Arnold, David, When Your Business Model Is Your Moat: The Case of Nest (2017) — HBS Case 9-617-051 - Hoffman, Patrick T., @arcktip Practitioner Framework: Systems Thinking for Hardware PMs (2026) — @arcktip - Blank, Steve, The Four Steps to the Epiphany, Introduction & Ch. 1 (2005) — customer development — works differently for hardware - Ulrich, Karl & Eppinger, Steven, Product Design and Development, Ch. 1–2 (2015) — McGraw-Hill — the engineering design process as PM context - Anderson, Chris, Makers: The New Industrial Revolution, Introduction & Ch. 1 (2012) — Crown — why the atom/bit divide is narrowing
2	Systems Thinking <i>Mapping firmware, cloud, connectivity, and UX as interdependent layers. Building a full dependency map for a physical-digital product.</i>	- Meadows, Donella, Thinking in Systems, Introduction & Ch. 1–2 (2008) — Chelsea Green — the most important conceptual tool in this course - Perlow, Leslie et al., Is Your Team Coordinating Too Much, or Not Enough? (2013) — HBR — dependency management in cross-functional teams - Hoffman, Patrick T., @arcktip Practitioner Framework: Dependency Mapping for Physical-Digital Products (2026) — @arcktip — the framework used in the Week 4 deliverable	- Meadows, Donella, Thinking in Systems, Introduction & Ch. 1–3 (2008) — Chelsea Green - Perlow, Leslie et al., Is Your Team Coordinating Too Much, or Not Enough? (2013) — HBR - Hoffman, Patrick T., @arcktip Practitioner Framework: Dependency Mapping for Physical-Digital Products (2026) — @arcktip - Senge, Peter, The Fifth Discipline, Introduction & Ch. 1–4 (1990) — Doubleday — systems archetypes and feedback loops - Bass, Len, Clements, Paul & Kazman, Rick, Software Architecture in Practice, Ch. 1–2 (2012) — Addison-Wesley — architecture for embedded systems context - Christensen, Clayton, The Innovator's Dilemma, Introduction & Ch. 1 (1997) — HBS Press — disruption most legible in hardware markets
3	Hardware Economics <i>Bill of Materials (BOM), yield rates, tariff exposure, and the volume pricing imperative. How hardware</i>	- DiResta, Renee, Forrest, Brady & Vinyard, Ryan, The Hardware Startup, Ch. 6 (Manufacturing) & Ch. 9 (Pricing) (2015) — O'Reilly — the most practical guide to hardware economics - Lutke, Tobias & Surowiecki, James, The Real Cost of Free	- DiResta, Renee, Forrest, Brady & Vinyard, Ryan, The Hardware Startup, Ch. 5–6 & 9–10 (2015) — O'Reilly - Chen, Elaine, Bringing a Hardware Product to Market, Ch. 3–5 (2015)

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	margin math differs fundamentally from SaaS.	Hardware (2015) — <i>Wired</i> — Juicero and the BOM trap — companion to the Week 14 case Chen, Elaine, Bringing a Hardware Product to Market, Ch. 4 (BOM & Costing) (2015) — Self-published — clear, practitioner-grade BOM walkthrough	- Lutke, Tobias & Surowiecki, James, The Real Cost of Free Hardware (2015) — <i>Wired</i> - Shapiro, Carl & Varian, Hal, Information Rules, Ch. 1–2 (1999) — HBS Press — switching costs and lock-in apply differently to hardware - Osterwalder, Alexander et al., Business Model Generation, Ch. 2 (Revenue Streams & Cost Structure) (2010) — Wiley — apply the canvas to a hardware product - Pisano, Gary, Profiting from Innovation in Drugs: Antecedents and Consequences of Alliance Formation in the Biotech Industry (1991) — <i>Strategic Management Journal</i> — regime of appropriability as hardware PM concept - Teece, David, Profiting from Technological Innovation, Ch. 1 (1986) — <i>Research Policy</i> — who captures value from hardware innovation?
4	The Product Roadmap How do you roadmap when a hardware change takes 18 months? Planning horizons, staged commitment, and the now/next/later framework adapted for hardware.	- Cagan, Marty, Inspired, Ch. 22–25 (Roadmaps & Product Vision) (2018) — read with hardware constraints in mind - Torres, Teresa, Continuous Discovery Habits, Introduction & Ch. 2 (2021) — <i>Product Talk</i> — opportunity mapping under hardware constraints - Hoffman, Patrick T., @arcktip Practitioner Framework: Roadmapping for Hardware Products (2026) — @arcktip — the framework adapted for the Week 4 deliverable	- Cagan, Marty, Inspired, Ch. 22–28 (2018) - Torres, Teresa, Continuous Discovery Habits, Introduction & Ch. 1–3 (2021) — <i>Product Talk</i> - Hoffman, Patrick T., @arcktip Practitioner Framework: Roadmapping for Hardware Products (2026) — @arcktip - Reinertsen, Donald, The Principles of Product Development Flow, Introduction & Ch. 1 (2009) — <i>Celeritas</i> — queuing theory applied to hardware development - Ward, Allen et al., The Second Toyota Paradox: How Delaying Decisions Can Make Better Cars Faster (1995) — <i>Sloan Management Review</i> — set-based design for hardware uncertainty - Wheelwright, Steven & Clark, Kim, Revolutionizing Product Development, Ch. 1 (1992) — <i>Free Press</i> — aggregate project planning for hardware portfolios
5	Case Study: The Copenhagen Wheel A full product lifecycle case: MIT Senseable City Lab origin, Superpedestrian commercialization, firmware complexity, B2C launch, and the pivot to B2B scooter fleets.	- Biderman, Assaf & Rus, Daniela, The Copenhagen Wheel: An Enabling Technology for Personal Mobility (2010) — MIT SENSEable City Lab — the original technical vision - Hoffman, Patrick T., @arcktip Practitioner Case: The Copenhagen Wheel Product Arc (2026) — @arcktip — practitioner reading on the lab-to-market journey - Chen, Elaine, Bringing a Hardware Product to Market, Ch. 7 (The Pivot) (2015) — product pivot under hardware constraints	- Biderman, Assaf & Rus, Daniela, The Copenhagen Wheel: An Enabling Technology for Personal Mobility (2010) — MIT SENSEable City Lab - Hoffman, Patrick T., @arcktip Practitioner Case: The Copenhagen Wheel Product Arc (2026) — @arcktip - Chen, Elaine, Bringing a Hardware Product to Market, Ch. 6–8 (2015) - Moore, Geoffrey, Crossing the Chasm, Ch. 2–3 (1991) — <i>HarperBusiness</i> — the chasm is wider for hardware

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			- Abernathy, William & Utterback, James, Patterns of Industrial Innovation (1978) — <i>Technology Review</i> — dominant design and hardware market transitions - Christensen, Clayton, The Innovator's Dilemma, Ch. 2–3 (1997) — <i>disruptive innovation from the hardware perspective</i>
6	Firmware & Software Integration OTA updates, backwards compatibility, rollback strategy, and the long tail of hardware in the field. The PM's guide to firmware as a live product.	- Margolis, Michael, Arduino Cookbook, Introduction (Embedded systems context) (2011) — O'Reilly — non-technical introduction to firmware constraints - Kleidermacher, David & Kleidermacher, Mike, Embedded Systems Security, Introduction (2012) — Elsevier — security as a firmware PM concern - Hoffman, Patrick T., @arcktip Practitioner Framework: Firmware and OTA Strategy for Hardware PMs (2026) — @arcktip — the practitioner's guide to firmware PM	- Margolis, Michael, Arduino Cookbook, Introduction & Ch. 1 (2011) — O'Reilly - Kleidermacher, David & Kleidermacher, Mike, Embedded Systems Security, Introduction & Ch. 1 (2012) — Elsevier - Hoffman, Patrick T., @arcktip Practitioner Framework: Firmware and OTA Strategy for Hardware PMs (2026) — @arcktip - Ganssle, Jack, The Art of Designing Embedded Systems, Introduction & Ch. 1 (2008) — Newnes — firmware engineering for PMs - Raymond, Eric S., The Cathedral and the Bazaar, Introduction & Essay 1 (1999) — O'Reilly — open source development model applied to firmware - Cusumano, Michael & Selby, Richard, Microsoft Secrets, Ch. 1 (1995) — Free Press — synchronize-and-stabilize for hardware-adjacent software
7	Supply Chain as Product Strategy Sole-source risk, near-shoring, component availability, and how supply chain decisions made in year one constrain every subsequent product decision.	- DiResta, Renee, Forrest, Brady & Vinyard, Ryan, The Hardware Startup, Ch. 6 (Supply Chain & Manufacturing) (2015) — O'Reilly - Huang, Andrew 'bunnie', The Essential Guide to Electronics in Shenzhen, Introduction & Ch. 1 (2016) — Sutamio Ko-Usagi — the canonical field guide to hardware sourcing - Chen, Elaine, Bringing a Hardware Product to Market, Ch. 5 (Suppliers & Partners) (2015)	- DiResta, Renee, Forrest, Brady & Vinyard, Ryan, The Hardware Startup, Ch. 6–7 (2015) - Huang, Andrew 'bunnie', The Essential Guide to Electronics in Shenzhen, Introduction & Ch. 1–2 (2016) — Sutamio Ko-Usagi - Chen, Elaine, Bringing a Hardware Product to Market, Ch. 5–6 (2015) - Fisher, Marshall, What Is the Right Supply Chain for Your Product? (1997) — HBR — functional vs. innovative products and supply chain design - Chopra, Sunil & Meindl, Peter, Supply Chain Management, Ch. 1–2 (2016) — Pearson — strategy and uncertainty in supply chains - Lee, Hau, The Triple-A Supply Chain (2004) — HBR — agile, adaptable, aligned — applies directly to hardware startups
8	Midterm: PRD Review Students present draft Product Requirements	Cagan, Marty, Inspired, Ch. 30–32 (Product Discovery & Spec) (2018) — the PM's standard for what a PRD must answer	- Cagan, Marty, Inspired, Ch. 30–35 (2018) - Ulrich, Karl & Eppinger, Steven, Product Design and

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	<i>Documents for a physical-digital product of their choosing. Structured critique on completeness, specificity, and feasibility.</i>	No new reading — prepare and present your PRD.	Development, Ch. 6 (Product Specifications) (2015) — McGraw-Hill — engineering specifications as PM discipline - No additional reading — prepare and present your PRD.
9	IoT & Connectivity MQTT, BLE, LTE-M, edge computing, data collection at scale — and the privacy and security implications of always-on hardware in the field.	- Greengard, Samuel, The Internet of Things, Introduction & Ch. 1–2 (2015) — MIT Press — accessible overview of IoT architecture - Schneier, Bruce, Click Here to Kill Everybody, Introduction & Ch. 1 (2018) — Norton — security implications of connected hardware - Hoffman, Patrick T., @arcktip Practitioner Framework: IoT Connectivity for Hardware PMs (2026) — @arcktip — protocols, edge vs. cloud, and fleet management	- Greengard, Samuel, The Internet of Things, Introduction & Ch. 1–3 (2015) — MIT Press - Schneier, Bruce, Click Here to Kill Everybody, Introduction & Ch. 1–3 (2018) — Norton - Hoffman, Patrick T., @arcktip Practitioner Framework: IoT Connectivity for Hardware PMs (2026) — @arcktip - Ashton, Kevin, How to Fly a Horse, Introduction & Ch. 1 (2015) — Doubleday — invention process grounding - Swan, Melanie, Blockchain: Blueprint for a New Economy, Introduction (2015) — O'Reilly — decentralized data architectures for IoT context - Zuboff, Shoshana, The Age of Surveillance Capitalism, Introduction (2019) — PublicAffairs — what IoT data collection enables at scale - Rose, David, Enchanted Objects, Introduction & Ch. 1 (2014) — Scribner — the design philosophy of ambient connected devices
10	Manufacturing Partners Working with contract manufacturers: NDA, NRE fees, EVT/DVT/PVT build cycles, acceptance criteria, and quality gate management.	- Chen, Elaine, Bringing a Hardware Product to Market, Ch. 8 (Contract Manufacturing) (2015) — the most practical chapter on CM relationships - Huang, Andrew 'bunnie', The Essential Guide to Electronics in Shenzhen, Ch. 3–4 (2016) — Sutajio Ko-Usagi — factory relationships in practice - DiResta, Renee, Forrest, Brady & Vinyard, Ryan, The Hardware Startup, Ch. 7 (Choosing a Manufacturer) (2015) — O'Reilly	- Chen, Elaine, Bringing a Hardware Product to Market, Ch. 7–9 (2015) - Huang, Andrew 'bunnie', The Essential Guide to Electronics in Shenzhen, Ch. 3–5 (2016) - DiResta, Renee, Forrest, Brady & Vinyard, Ryan, The Hardware Startup, Ch. 7–8 (2015) - Pisano, Gary & Shih, Willy, Restoring American Competitiveness (2009) — HBR — industrial commons and what happens when manufacturing leaves - Utterback, James, Mastering the Dynamics of Innovation, Ch. 5 (1994) — HBS Press — process innovation and manufacturing maturity - Sturgeon, Timothy, Modular Production Networks: A New American Model of Industrial Organization (2002) — Industrial and Corporate Change — contract manufacturing as industrial system
11	Field Reliability & Support	Hoffman, Patrick T., @arcktip Course Template: Hardware Economics Model (2026) — @arcktip — the Week 11	Hoffman, Patrick T., @arcktip Course Template: Hardware Economics Model (2026) — @arcktip

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	What happens after ship: failure modes, MTBF, field returns analysis, firmware rollbacks, and running customer support for a hardware product at scale.	deliverable template; read and prepare inputs - Mocker, Monica & Arnold, David, When Your Business Model Is Your Moat: The Case of Nest (2017) — HBS Case — return to the Nest case with field reliability lens - Chen, Elaine, Bringing a Hardware Product to Market, Ch. 10 (After Launch) (2015)	- Chen, Elaine, Bringing a Hardware Product to Market, Ch. 10–11 (2015) - Juran, J.M. & Godfrey, A. Blanton, Juran's Quality Handbook, Introduction & Ch. 1 (1998) — McGraw-Hill — quality engineering concepts for PMs - Kano, Noriaki et al., Attractive Quality and Must-Be Quality (1984) — <i>Journal of the Japanese Society for Quality Control</i> — the Kano model for hardware reliability - Reinertsen, Donald, The Principles of Product Development Flow, Ch. 6 (Managing Queues) (2009) — Celeritas — managing field support as a queue problem - MacKenzie, Donald, Inventing Accuracy, Introduction (1990) — MIT Press — reliability and precision as socially constructed standards
12	Enterprise Hardware B2B physical-digital products: enterprise procurement cycles, IT security requirements, fleet management, and what changes when your buyer is an IT department.	- Cagan, Marty, Inspired, Ch. 4–5 (Enterprise Companies) (2018) — enterprise PM vs. consumer PM — the structural differences - Dixon, Matthew, Adamson, Brent & Toman, Nicholas, The Challenger Sale, Introduction & Ch. 1 (2011) — Portfolio — B2B sales as a product PM concern - Hoffman, Patrick T., @arcktip Practitioner Framework: Enterprise Hardware PM (2026) — @arcktip — procurement cycles, RFPs, and fleet dashboards	- Cagan, Marty, Inspired, Ch. 4–5 & 15–18 (2018) - Dixon, Matthew, Adamson, Brent & Toman, Nicholas, The Challenger Sale, Introduction & Ch. 1–2 (2011) — Portfolio - Hoffman, Patrick T., @arcktip Practitioner Framework: Enterprise Hardware PM (2026) — @arcktip - Foote, David, IT Procurement Cycles for Hardware: A Practitioner's Guide (2020) — Gartner — enterprise hardware procurement reality - Shapiro, Benson & Moriarty, Rowland, Support Systems for Industrial Selling, Ch. 1 (1984) — HBS — B2B sales support as PM responsibility - Utterback, James, Mastering the Dynamics of Innovation, Ch. 7 (Dominant Design in B2B) (1994) — HBS Press
13	Platform Strategy When does hardware become a platform? Lock-in mechanics, API strategy, the razor-and-blades model, and how Apple, Peloton, and JUMP built platform businesses on physical products.	- Parker, Geoffrey, Van Alstyne, Marshall & Choudary, Sangeet, Platform Revolution, Introduction & Ch. 1 (2016) — Norton — platform mechanics, starting with physical examples - Shapiro, Carl & Varian, Hal, Information Rules, Ch. 7 (Networks and Positive Feedback) (1999) — HBS Press — lock-in and switching costs in hardware platforms - Hoffman, Patrick T., @arcktip Practitioner Framework: Pricing as Product Design (2026) — @arcktip — platform pricing in a physical product context	- Parker, Geoffrey, Van Alstyne, Marshall & Choudary, Sangeet, Platform Revolution, Introduction & Ch. 1–3 (2016) — Norton - Shapiro, Carl & Varian, Hal, Information Rules, Ch. 5–7 (1999) — HBS Press - Hoffman, Patrick T., @arcktip Practitioner Framework: Pricing as Product Design (2026) — @arcktip - Cusumano, Michael & Gawer, Annabelle, The Elements of Platform Leadership (2002) — MIT Sloan Management Review — platform architecture decisions - Gawer, Annabelle & Cusumano, Michael, Platform Leadership, Introduction & Ch. 1 (2002) — HBS Press — Intel, Cisco, and the platform PM role

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			• Rochet, Jean-Charles & Tirole, Jean, Platform Competition in Two-Sided Markets (2003) — <i>Journal of the European Economic Association</i> — two-sided market theory • Zhu, Feng & Iansiti, Marco, Why Some Platforms Thrive and Others Don't (2019) — HBR — platform success factors including hardware-based ones
14	Sunset & Obsolescence <i>End-of-life planning, e-waste obligations, hardware recycling regulation, right-to-repair, and what 'done' means when your product has atoms.</i>	• Crawford, Kate, Atlas of AI, Ch. 1 (Earth) (2021) — Yale UP — the material supply chain and e-waste reality of hardware • Perzanowski, Aaron & Schultz, Jason, The End of Ownership, Introduction & Ch. 1 (2016) — MIT Press — right-to-repair and hardware ownership law • EU Right to Repair Directive (2023) — Executive Summary	• Crawford, Kate, Atlas of AI, Ch. 1–2 (2021) — Yale UP • Perzanowski, Aaron & Schultz, Jason, The End of Ownership, Introduction & Ch. 1–3 (2016) — MIT Press • EU Right to Repair Directive (2023) — Executive Summary • Stahel, Walter, The Circular Economy: A User's Guide, Introduction & Ch. 1 (2019) — Routledge — circular economy frameworks for product EOL • MacKenzie, Donald, Inventing Accuracy, Ch. 6 (Standards and Obsolescence) (1990) — MIT Press — how standards shape hardware lifetimes • Agrawal, Rakesh, E-Waste: Management, Processing, and Technology (2020) — Elsevier — regulatory and technical dimensions of hardware disposal • Mazzucato, Mariana, The Entrepreneurial State, Ch. 4 (2013) — Anthem — public investment in green hardware infrastructure
15	Final Pitch <i>Teams present a complete product concept: PRD, roadmap, BOM estimate, dependency map, supply chain strategy, and go-to-market plan — to a panel of hardware practitioners.</i>	• No assigned reading — final pitch preparation only.	• No assigned reading — final pitch day. • Hoffman, Patrick T., @arcktip Synthesis: Hardware Product Strategy (2026) — @arcktip — optional synthesis essay; the practitioner's checklist

Reading Notes

Titles in *italic* are assigned texts. 'Ch.' indicates specific chapters only. HBS Cases available via Harvard Business Publishing. All other readings available via university library or open access. @arcktip sources (Hoffman) are primary practitioner readings — first-hand frameworks from building Social Bicycles / JUMP (physical-digital mobility products) and product leadership at Google and Meta. The Copenhagen Wheel case (Week 5) draws directly on Superpedestrian's product arc.

This is a skills-first course. Readings are selected to equip students to do hardware PM work — BOM modeling, dependency mapping, PRD writing, firmware management — not primarily to analyze hardware products from the outside.